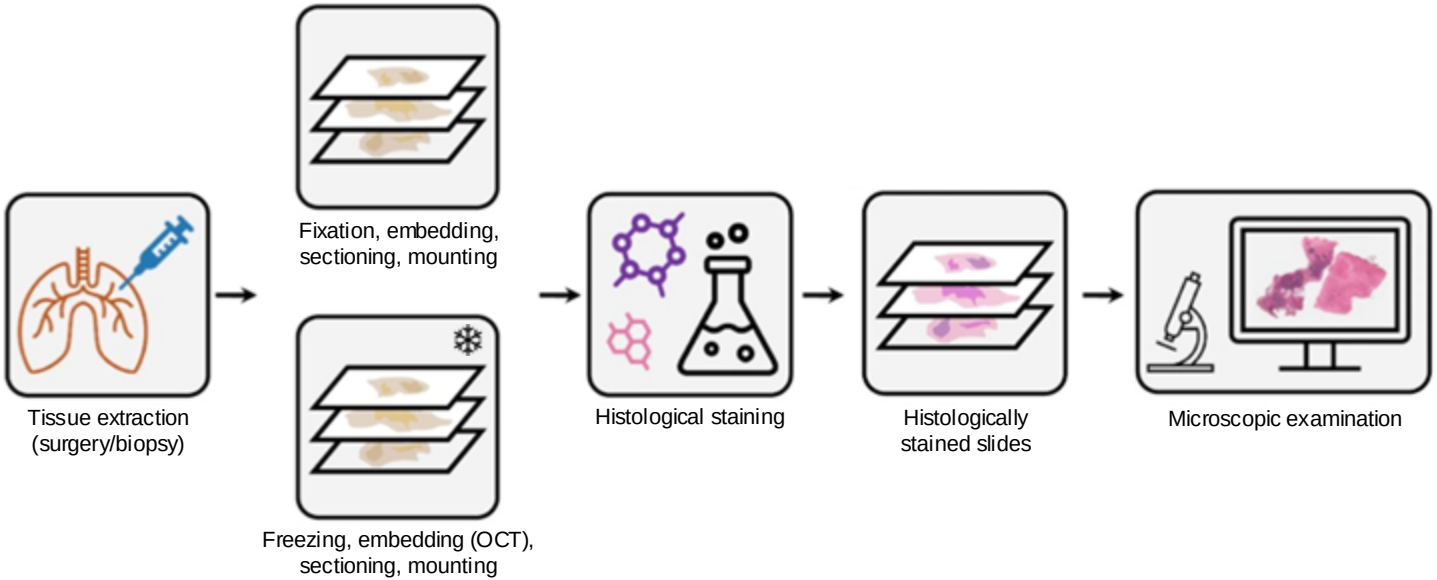
Representation Learning and Data-Centric Approaches in Computational Pathology

Instantiation to Alzheimer's Disease

Gabriel JIMENEZ
Supervised by Prof. Daniel RACOCEANU

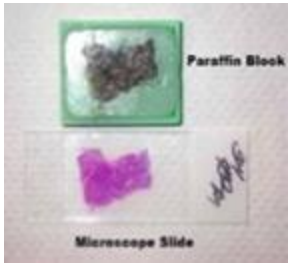
18/10/2024

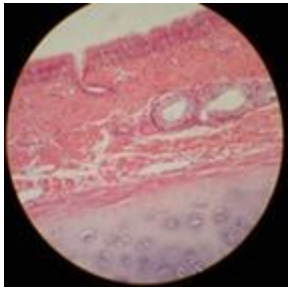
The standard histological staining process



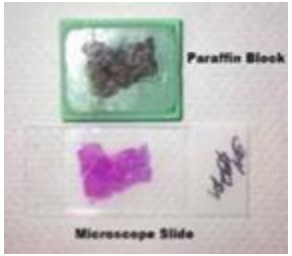
Bai, B., Yang, X., Li, Y. et al. Deep learning-enabled virtual histological staining of biological samples. Light Sci Appl 12, 57 (2023)

Histopathology → Digital Pathology



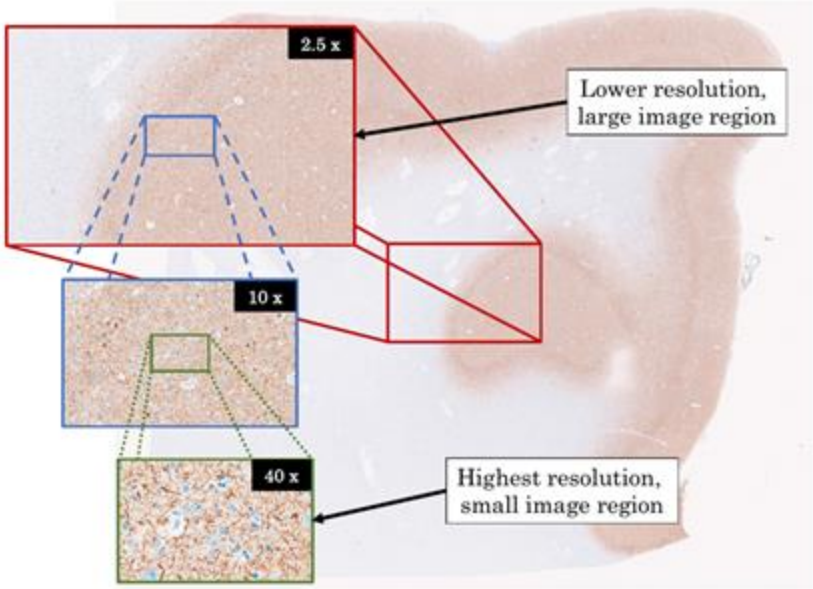
Microscopic examination of a biopsy or surgical specimen by a pathologist



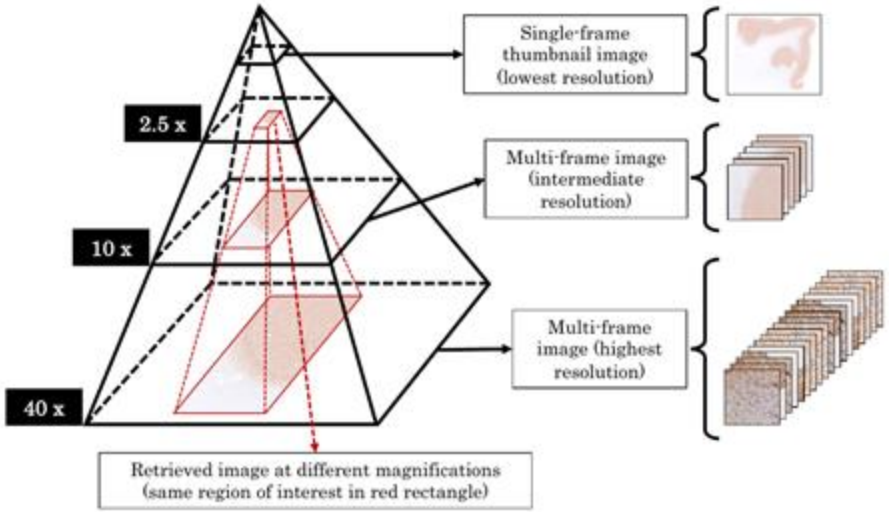



The management and interpretation of pathology information generated from a digitized glass slide

Whole Slide Images



Rapid zooming issue when accessing lower resolution images



Pyramidal organization of Whole Slide Images

Jimenez, G., Racoceanu, D. (2023). Computational Pathology for Brain Disorders. In: Colliot, O. (eds) Machine Learning for Brain Disorders. Neuromethods, vol 197. Humana, New York, NY.

Mechanical and hardware related errors



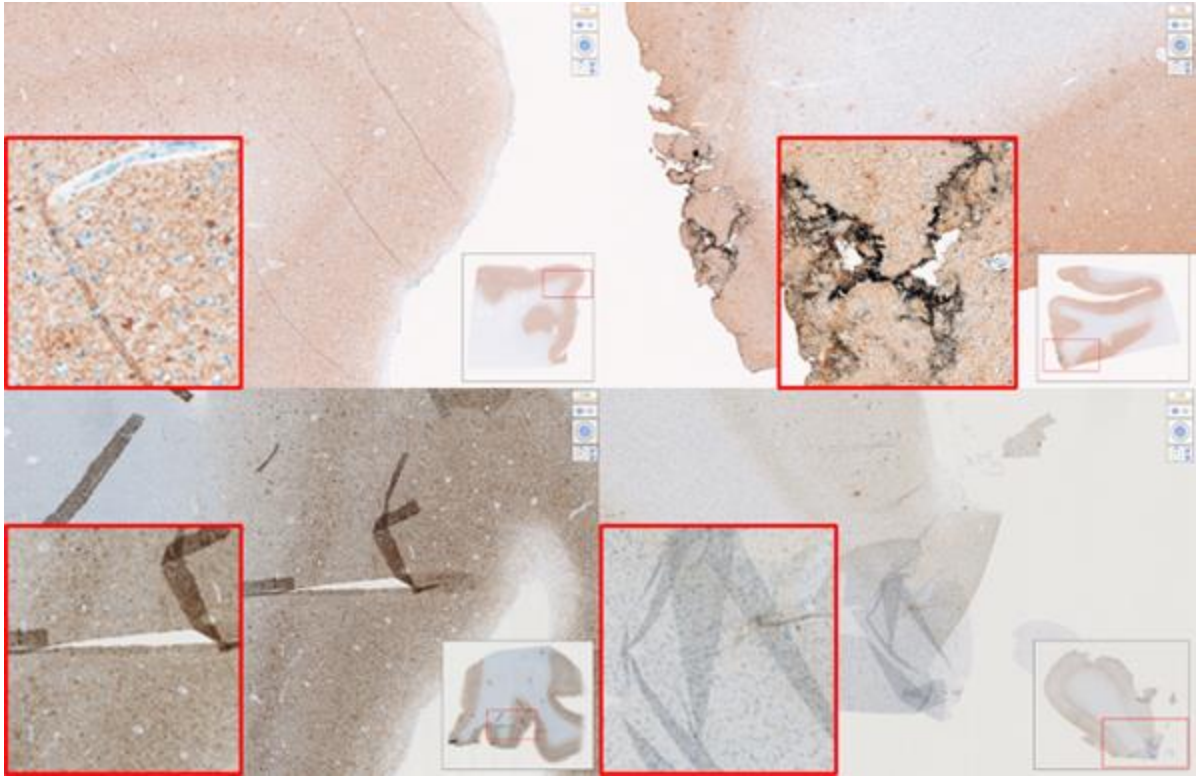


Every WSI scanner has six components: (a) a microscope with **lens objectives**, (b) **light source** (bright field and/or fluorescent), (c) **robotics** to load and move glass slides around, (d) one or more **digital cameras** for capture, (e) a **computer**, and (f) **software** to manipulate, manage, and view digital slides

Imaging artifacts

Folding artifact (floatation and mounting related artifact)

Marking fixation process (fixation artifact)



Breaking artifact (microtome-related artifact)

Overlaying tissue (mounting artifact)

*Jiménez, G., Racoceanu, D. (2023). Computational Pathology for Brain Disorders. In: Colliot, O. (eds) Machine Learning for Brain Disorders. Neuromethods, vol 197. Humana, New York, NY.

StratifIAD project [BBT3]

Interpretable Deep Learning in Computational Histopathology for Alzheimer Disease Patient Stratification Refinement

Gabriel JIMENEZ, **Daniel RACOCEANU**
Alzheimer's disease and prion diseases (ICM): Lev STIMMER, **Benoit DELATOUR**

18/10/2024

Understanding Alzheimer's Disease

Alzheimer disease is caused by

- Misfolding and accumulation of Amyloid-beta peptides and **tau proteins.**

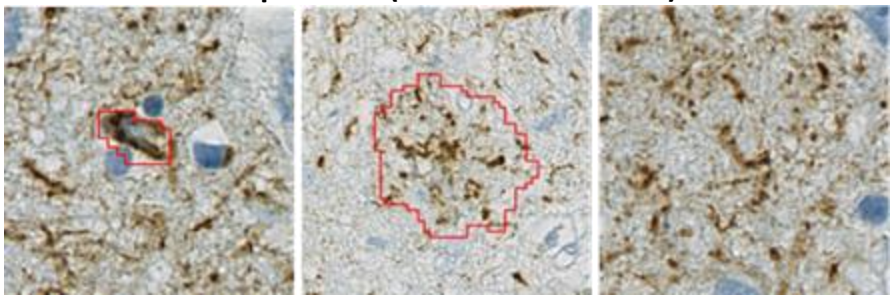
Findings by Duyckaerts, et al.*:

- Tau pathologies (tauopathies) are highly linked to clinical manifestations of AD.

The problem:

- The clinical presentation of the patients is very heterogeneous.
- In vivo imaging methodologies are not able to reach the resolution obtained by microscopy.

Tau proteins (located in neurons)



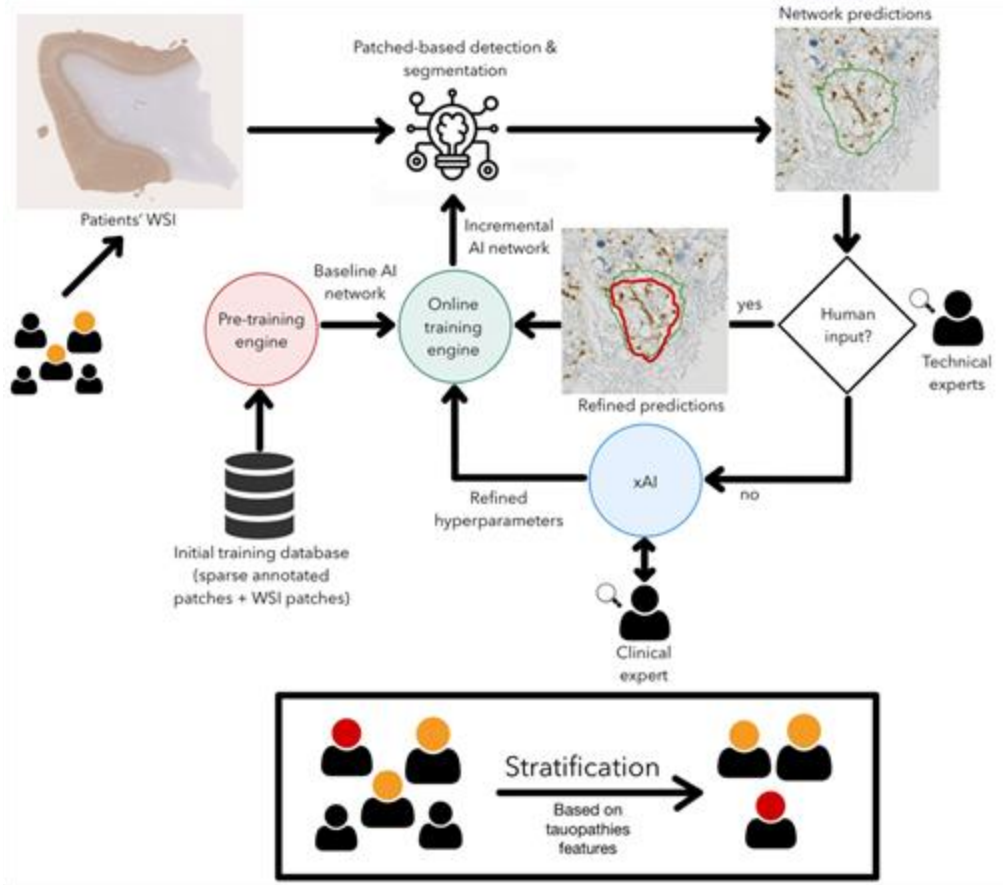
a. Neurofibrillary tangle **b. Neuritic Plaque** **c. Dystrophic Neurites**

*Duyckaerts C., Delatour B., Potier MC. Classification and basic pathology of Alzheimer disease. Acta Neuropathol. 2009.

Annotation Refinement Tool

- Automatic **segmentation/detection** of tau protein aggregates from WSI.

- Manual annotation refinement using DL attention mechanisms and visual explainability features from DL models.

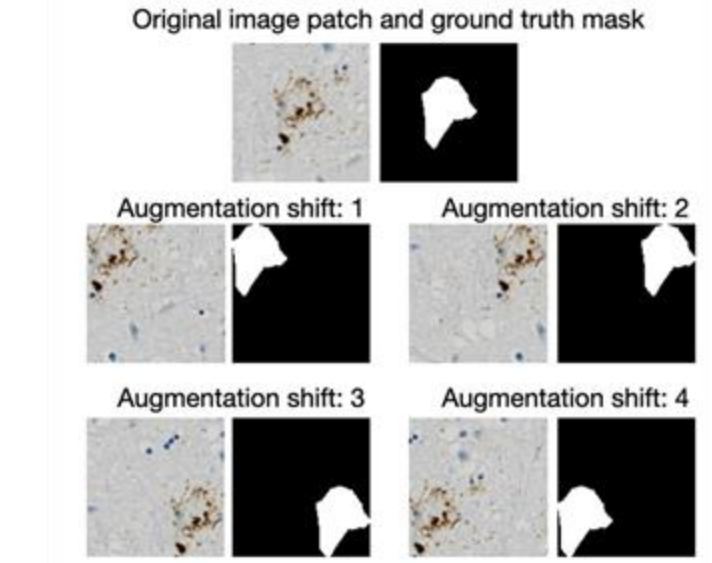


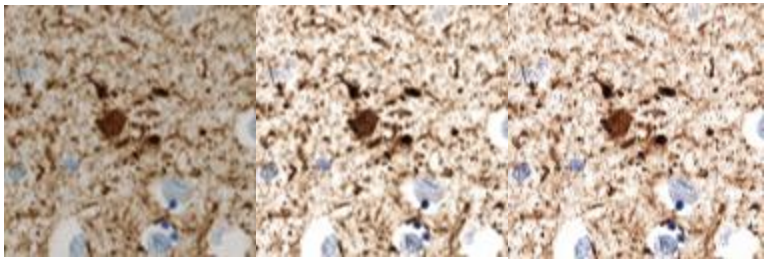
cytomine

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Brain WSI datasets

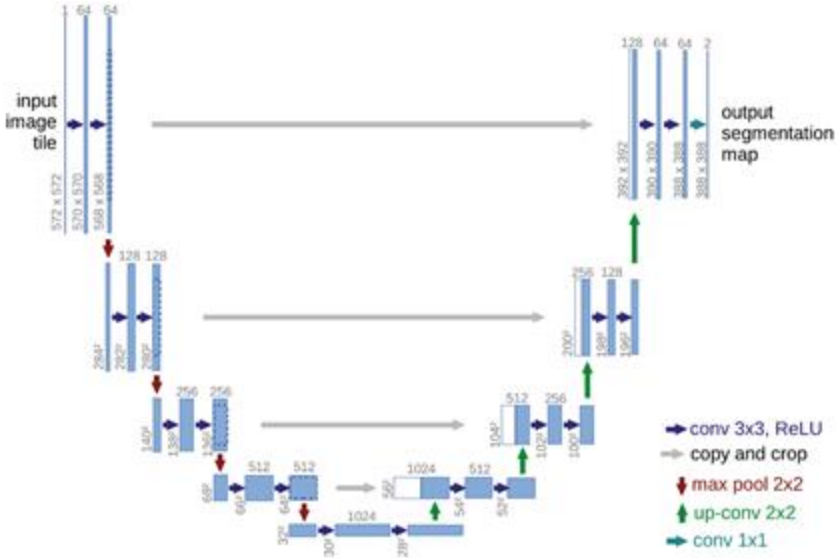
ALZ50 dataset	AT8 dataset
~3K patches (6 WSI)	~16K patches (8 WSI)
NanoZoomer 2.0-RS	NanoZoomer 2.0-RS & S60
ALZ50 antibodies	AT8 antibodies
Manual annotation	Manual annotation
BRS & EDS	ROI-guided sampling
Plaques (NP)	Plaques (NP)
128x128	128x128 & 256x256
20x (plaques) 227 nm/px @ 40x	20x (plaques) 221 nm/px @ 40x



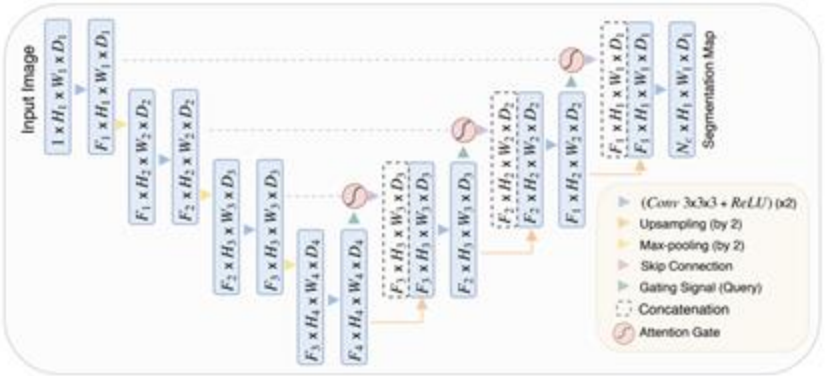


Original Macenko Vahadane

Deep learning architectures



Olaf R., et al. U-Net: Convolutional Networks for Biomedical Image Segmentation. MICCAI, 2015



Ozan O., et al. Attention U-Net: Learning Where to Look for the Pancreas. MIDL, 2018

- **ALZ50 dataset: detection and segmentation. Architecture: UNet[1]**

Task	Normalization	F1 (cross-val)	F1 (cross-test)
Det.	Macenko	99.8 %	81.3 %
Seg.	Vahadane LS	81.5 %	78.2 %

- **AT8 dataset: segmentation results[2]**

Architecture: UNet

Patch size	Normalization	F1 (cross-val)	F1 (cross-test)
128x128	Macenko	69.5 ± 2.9 %	68.5 ± 2.6 %
256x256	Macenko	66.0 ± 4.2 %	64.6 ± 3.3 %

Architecture: Attention UNet

Patch size	Normalization	F1 (cross-val)	F1 (cross-test)
128x128	Macenko	75.2 ± 3.3 %	69.20 ± 2.5 %
256x256	Macenko	69.31 ± 4.5 %	63.42 ± 3.0 %

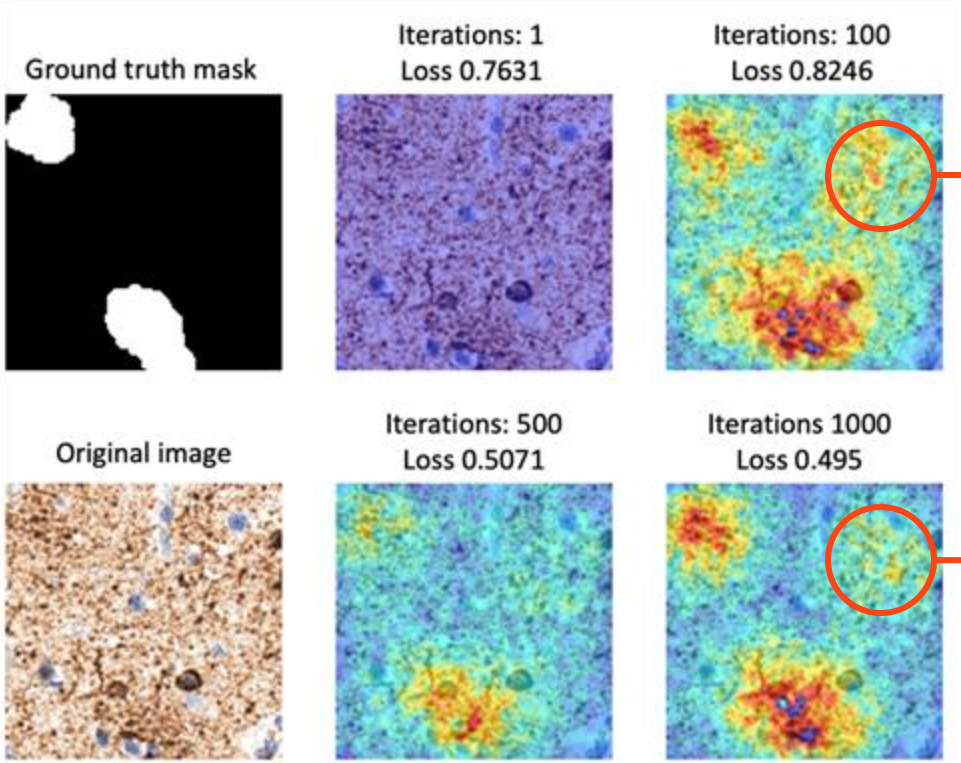
- **AT8 dataset: different scanners & normalization techniques. Architecture: UNet[2]**

Scanner	Patch size	Normalization	F1 (cross-val)
NanoZoomer 2.0-RS	128x128	Macenko	73.4 ± 0.6 %
NanoZoomer S60	128x128	Macenko	63.5 ± 2.4 %

Normalization	Patch size	F1 (test)
Macenko	128x128	72.5 %
Vahadane	128x128	70.9 %

[1]Manouskova, K., Abadie, V., Ounissi, M., Jimenez, G., Stimmer, L., Delatour, B., Durrleman, S., Racoceanu, D. – SPIE Medical Imaging 2022
[2]Jimenez, G., Kar, A., Ounissi, M., Ingrassia, L., Boluda, S., Delatour, B., Stimmer, L., Racoceanu, D. – MICCAI 2022

- **Visual explanation properties of DL model → improving annotations**



Conclusion

a. Antibodies impact the detection and segmentation of tau aggregates in WSI.
   - ALZ50 → misfolded tau proteins
   - AT8 → phosphorylated tau proteins

a. Visual explainability to improve manual annotations.
   - Improve morphological analysis of tau aggregates.

a. Benchmark with commercial software
   - Outperformed in test dataset. Commercial software follows a black-box approach → no explainability

d. High performance variability using **different scanners**



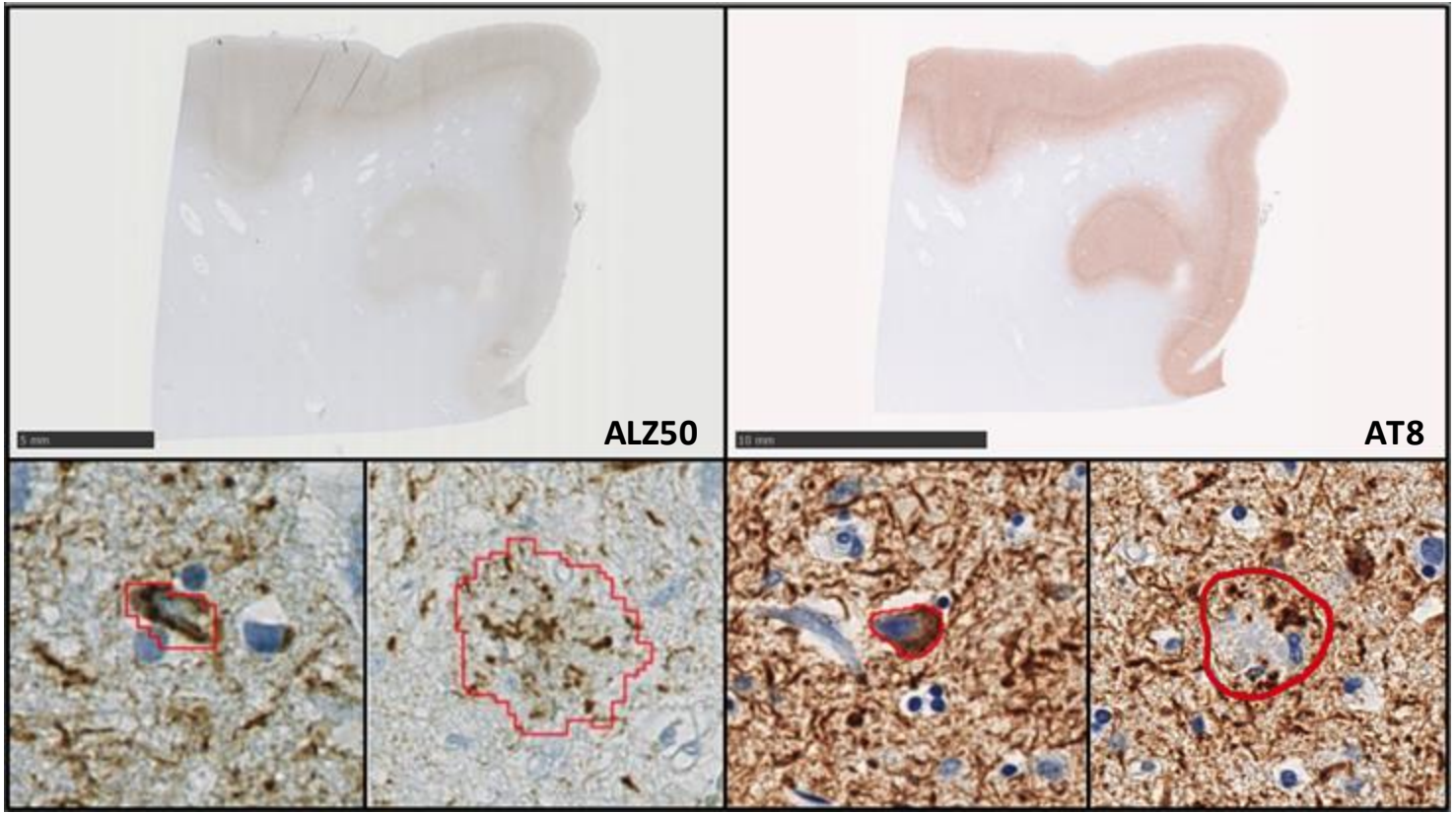
e. Performance variability due to **patch size**

Frequency representation of WSI

Normalization of WSI

18/10/2024

Staining differences in WSI



Color vs. Frequency representation of images

	Color representation	Frequency representation
Noise sensitivity	✗	✓
Details in homogeneous regions	!	✓
Structural information	↓	↑
Detection of low-contrast regions	!	↑
Data compression	✗	✓

Amplitude-Frequency demodulation of images

$$I(x, y) = \sum_{n=1}^{n=M} a(x, y) \cos(n\, \varphi(x, y))$$

$$\nabla\varphi(x, y) = \left(\frac{\partial \varphi}{\partial x}(x, y), \frac{\partial \varphi}{\partial y}(x, y) \right)$$

$$f_{AS}(x, y) = f(x, y) + j\mathcal{H}_{2d}[f(x, y)],$$

$$a(x, y) = |f_{AS}(x, y)|,$$

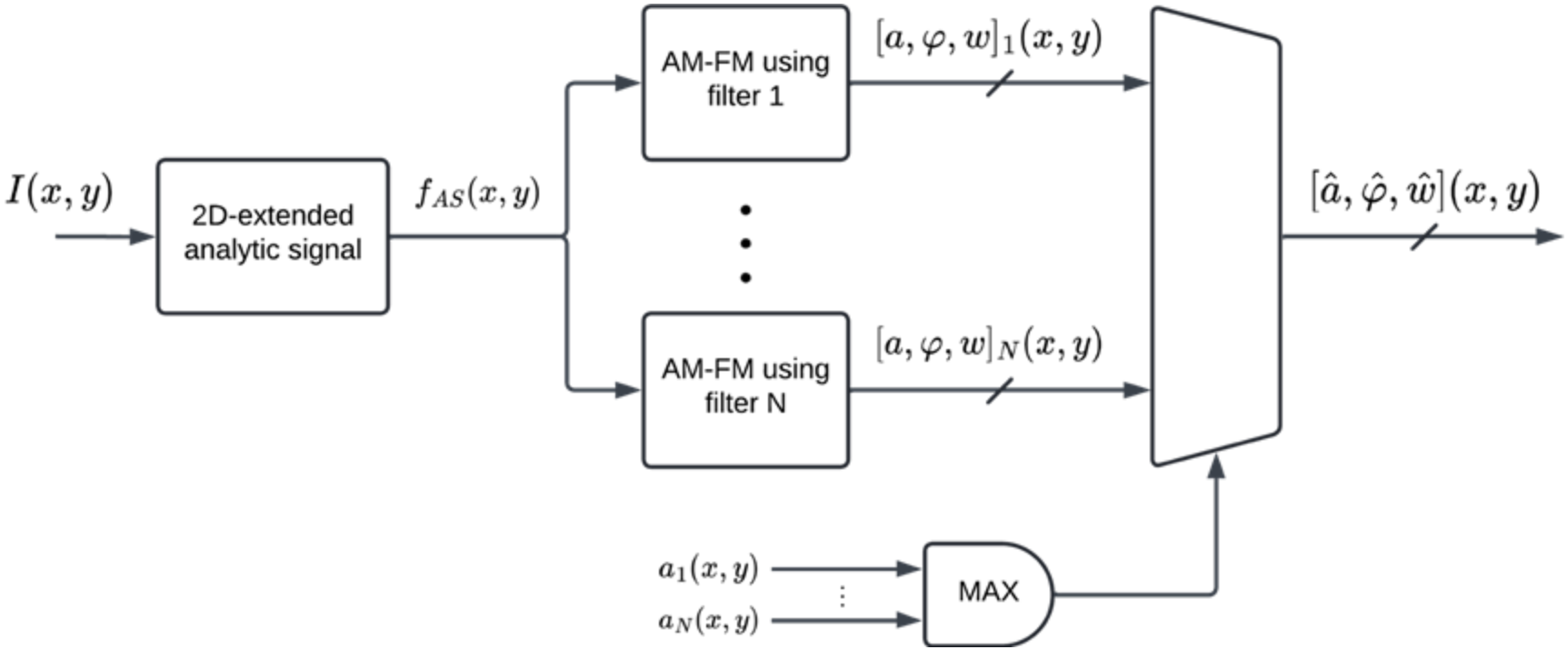
$$\varphi(x, y) = \arctan\left(\frac{\text{imag}(f_{AS}(x, y))}{\text{real}(f_{AS}(x, y))} \right)$$

$$\omega(x, y) = \text{real}\left[-j \frac{\nabla f_{AS}(x, y)}{f_{AS}(x, y)} \right].$$

*Murray, Victor, Paul Rodriguez, and Marios S. Pattichis. 2010. "Multiscale AM-FM Demodulation and Image Reconstruction Methods with Improved Accuracy." IEEE Transactions on Image Processing: A Publication of the IEEE Signal Processing Society 19 (5): 1138–52.

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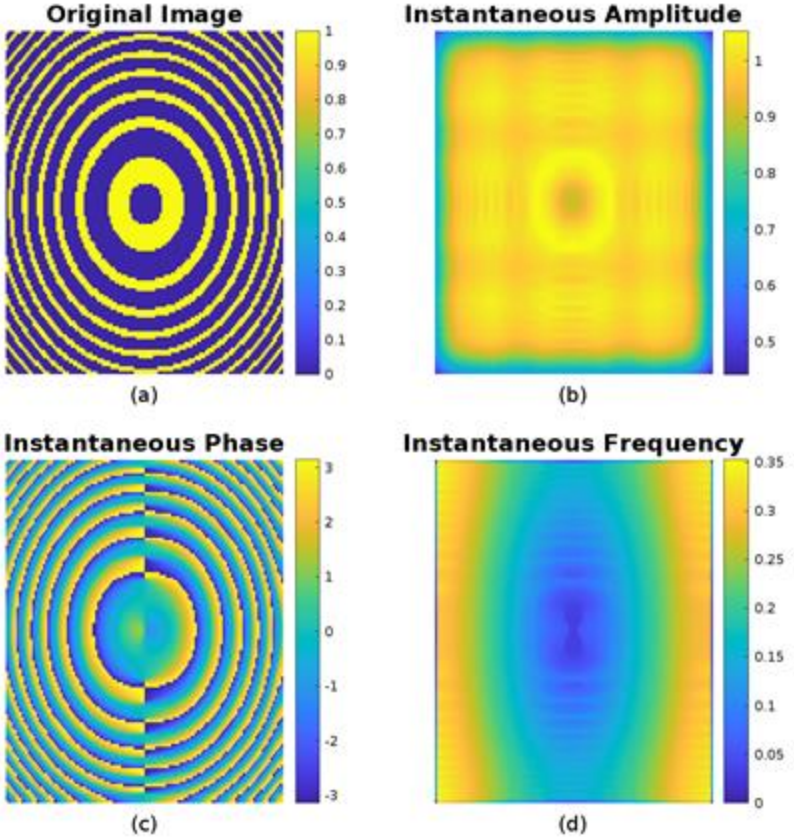
Amplitude-Frequency demodulation of images

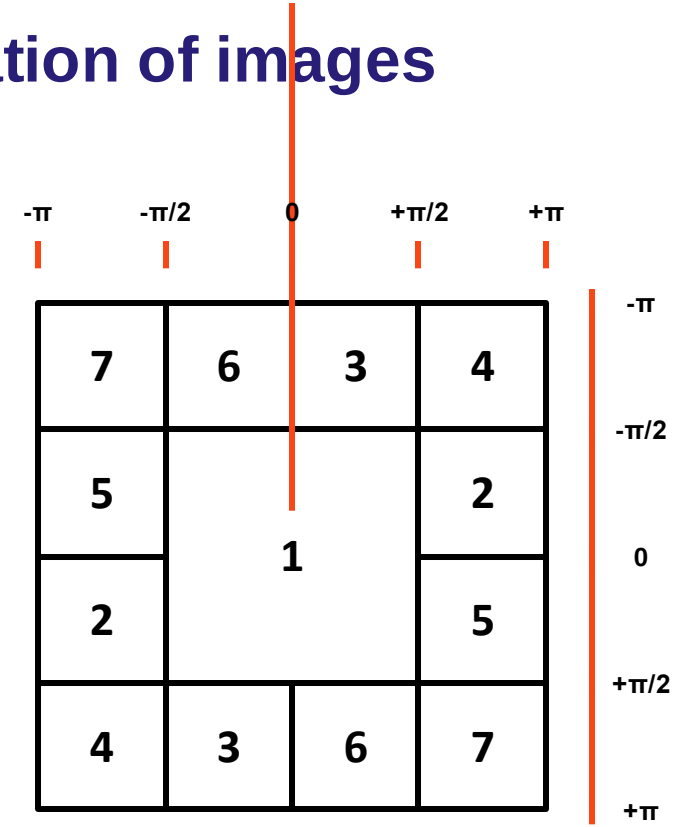


Murray, Victor, Paul Rodriguez, and Marios S. Pattichis. 2010. "Multiscale AM-FM Demodulation and Image Reconstruction Methods with Improved Accuracy." IEEE Transactions on Image Processing: A Publication of the IEEE Signal Processing Society 19 (5): 1138–52.

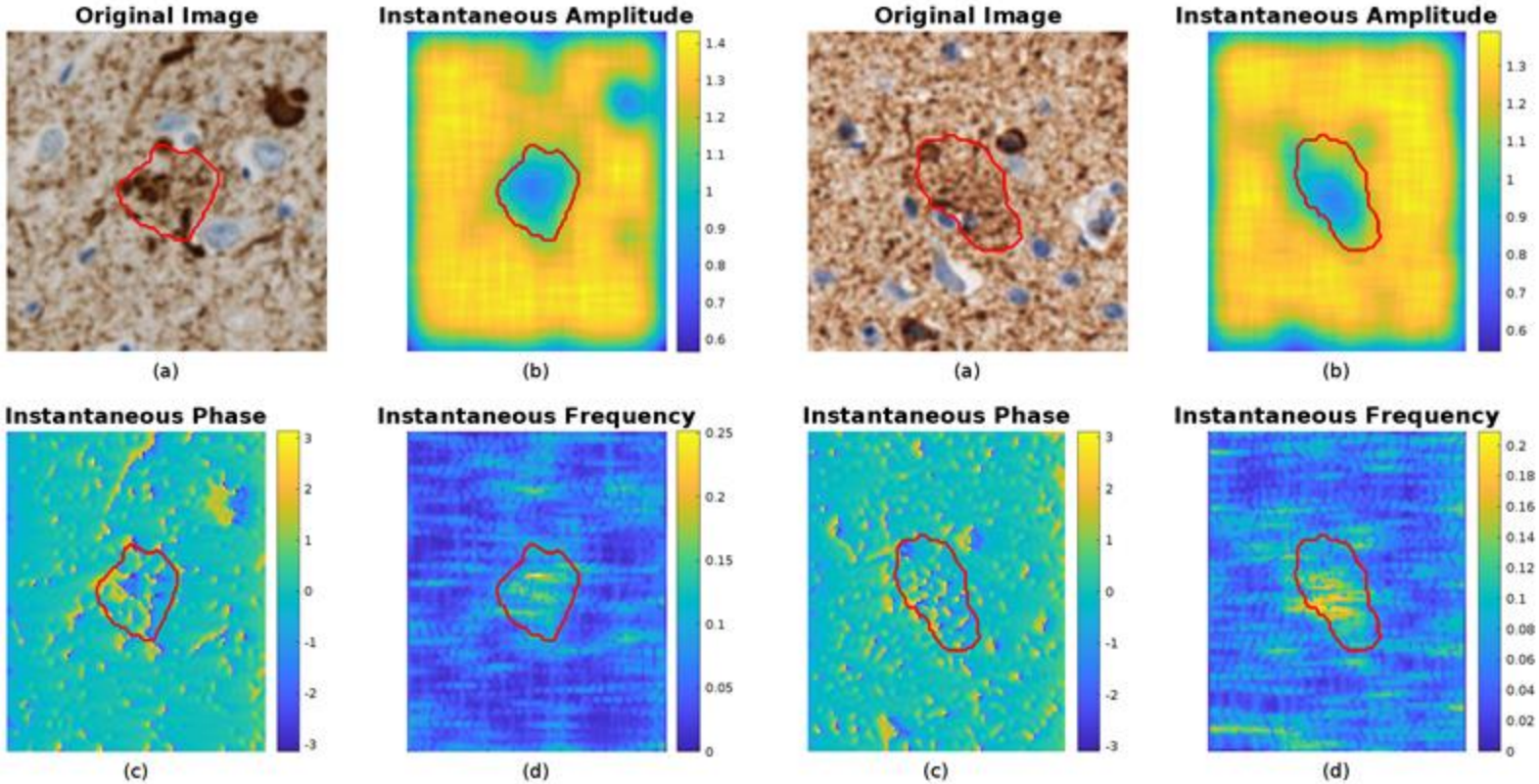
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Amplitude-Frequency demodulation of images





Amplitude-Frequency demodulation of WSI



NanoZoomer 2.0-RS **[DS: 0.73 (±0.03)]**

NanoZoomer S60 **[DS: 0.74 (±0.02)]**

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Conclusion

a. No significant variability using different scanners
   - NanoZoomer 2.0-RS → 73.4 ± 0.6 % [Macenko] vs. 72.7 ± 2.6 % [AM-FM demodulation]
   - NanoZoomer S60 → 63.5 ± 2.4 % [Macenko] vs. 74.1 ± 1.5 % [AM-FM demodulation]

a. No need for reference image.
   - Bias in the selection of the most representative sample in the dataset.

a. Frequency-normalized images are bounded by definition.
   - IP and IF are bounded to the normalized fundamental frequency range.

d. Filter design.

e. **Patch size** influence.



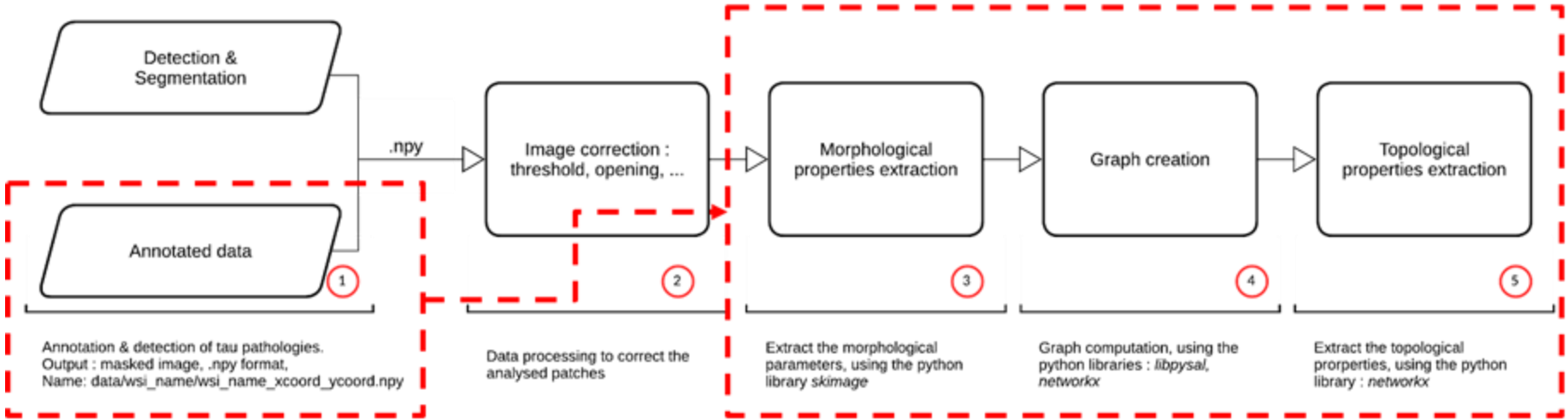
e. Multiple scales for the IA, IP, IF estimation.

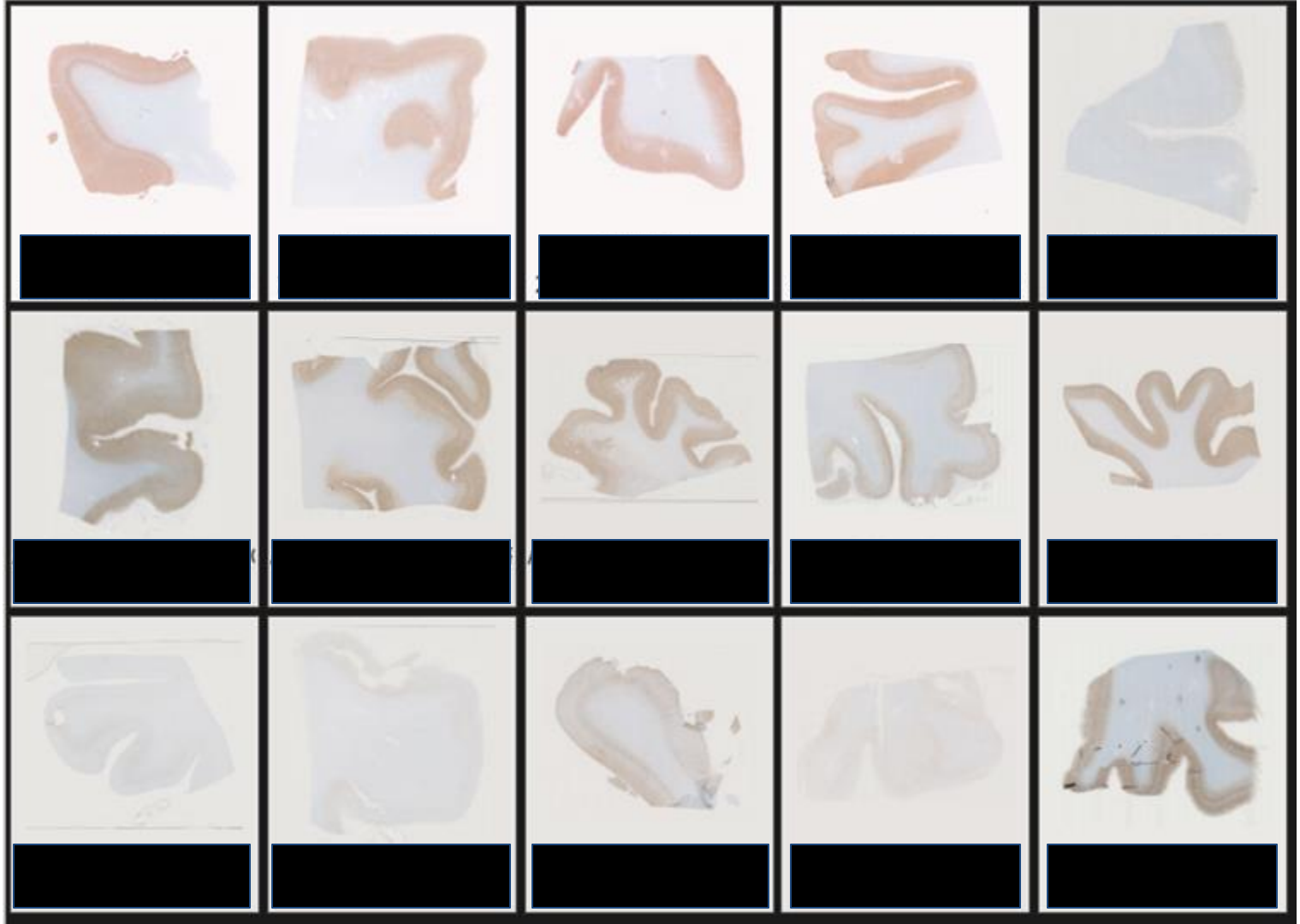
Graph representation of WSI

Analyzing local and global patterns in WSI

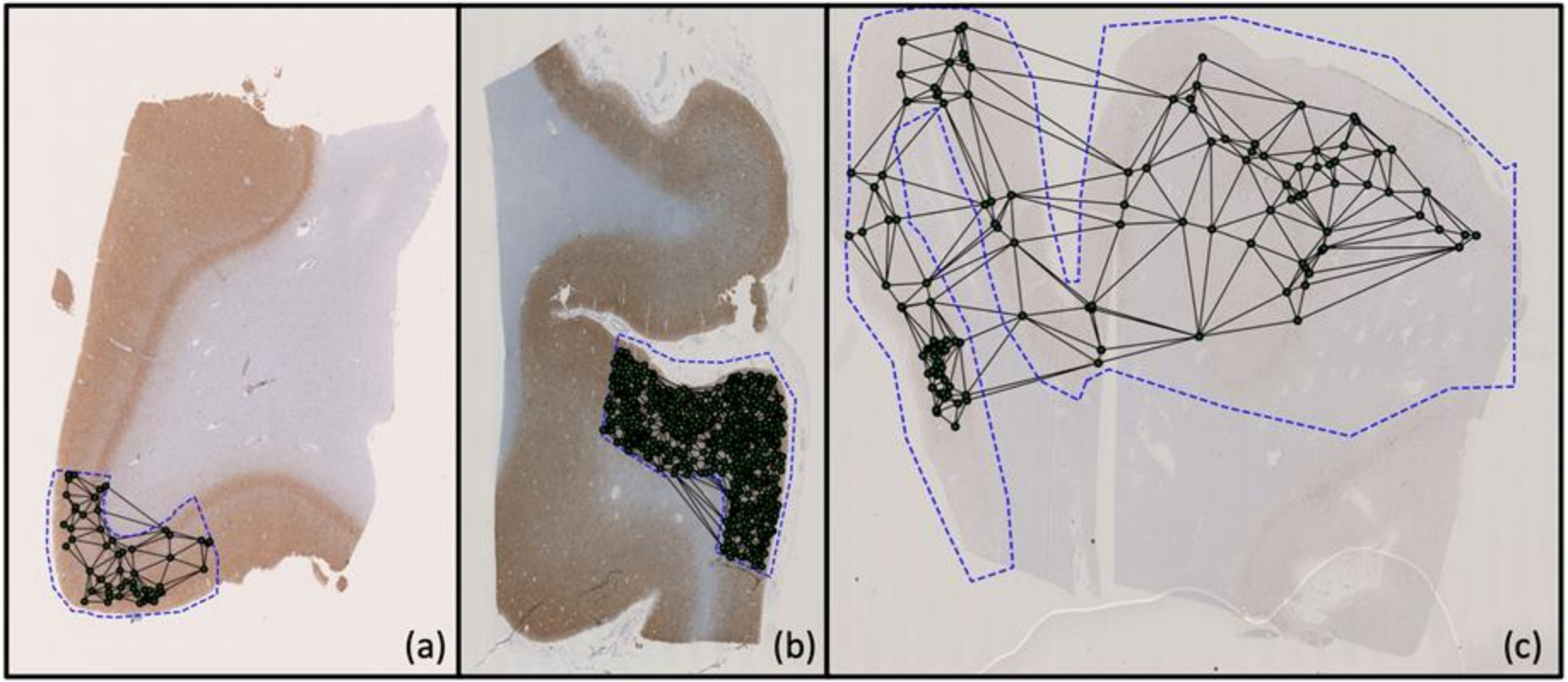
18/10/2024

What about global information of the WSI?

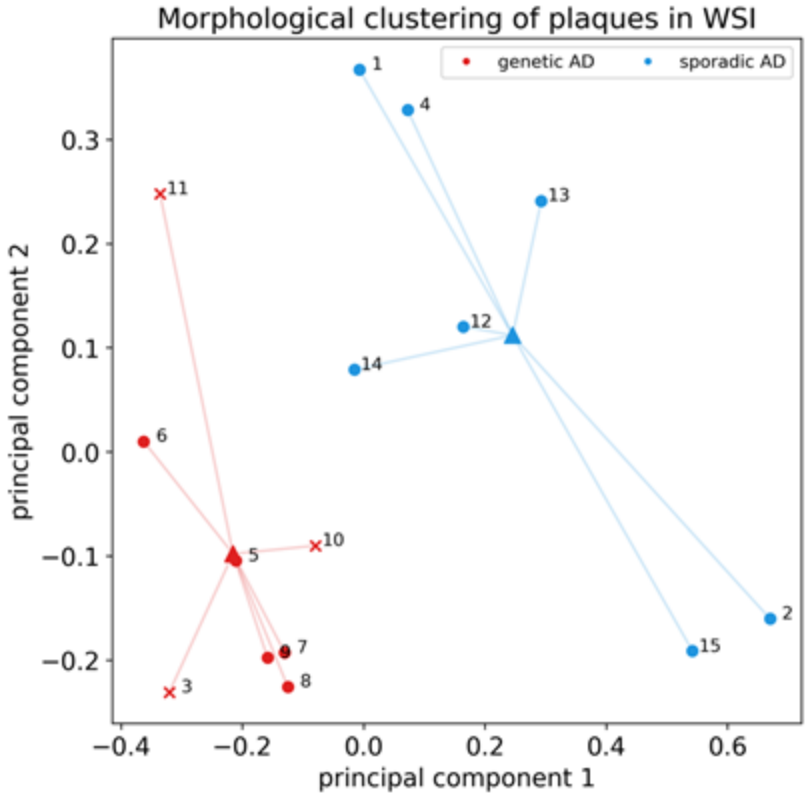


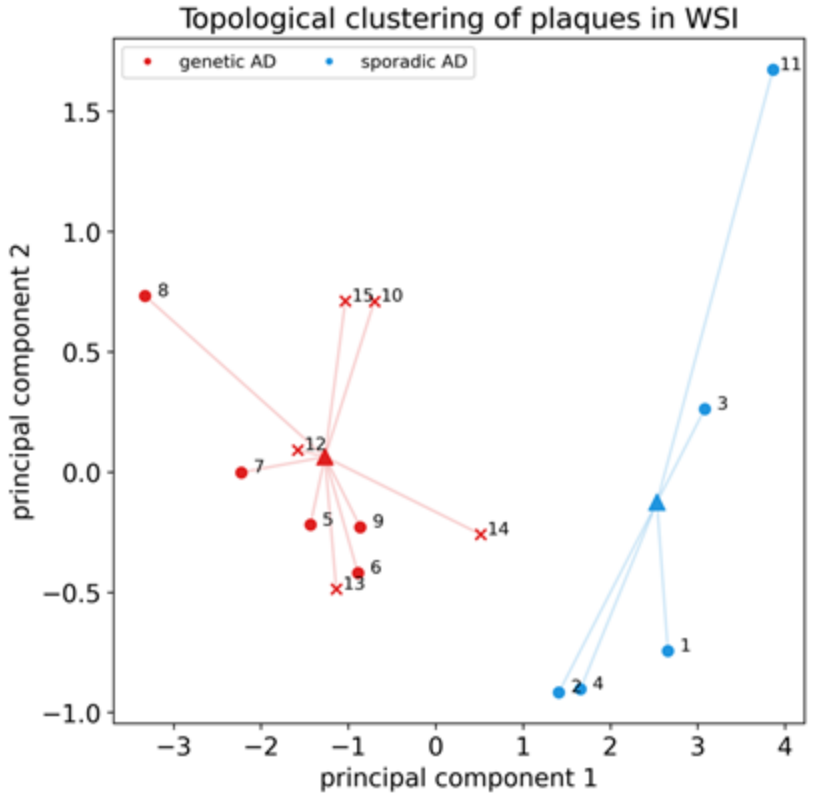


Graph creation: one WSI/graph per patient



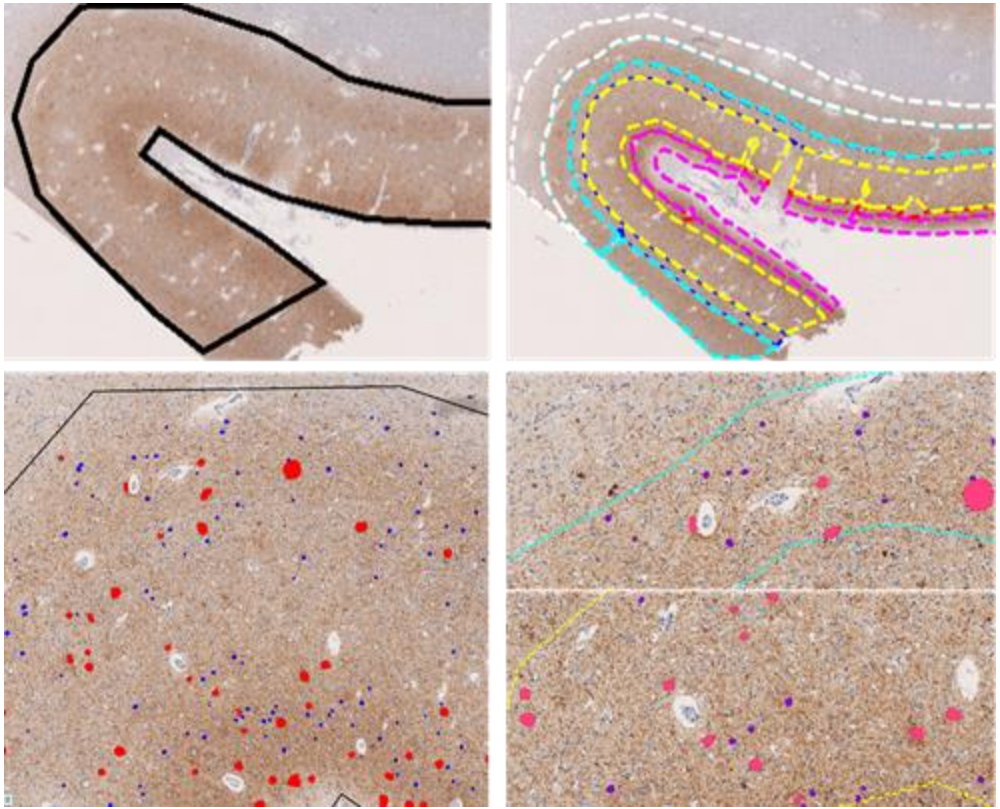
Clustering: morphological and topological features





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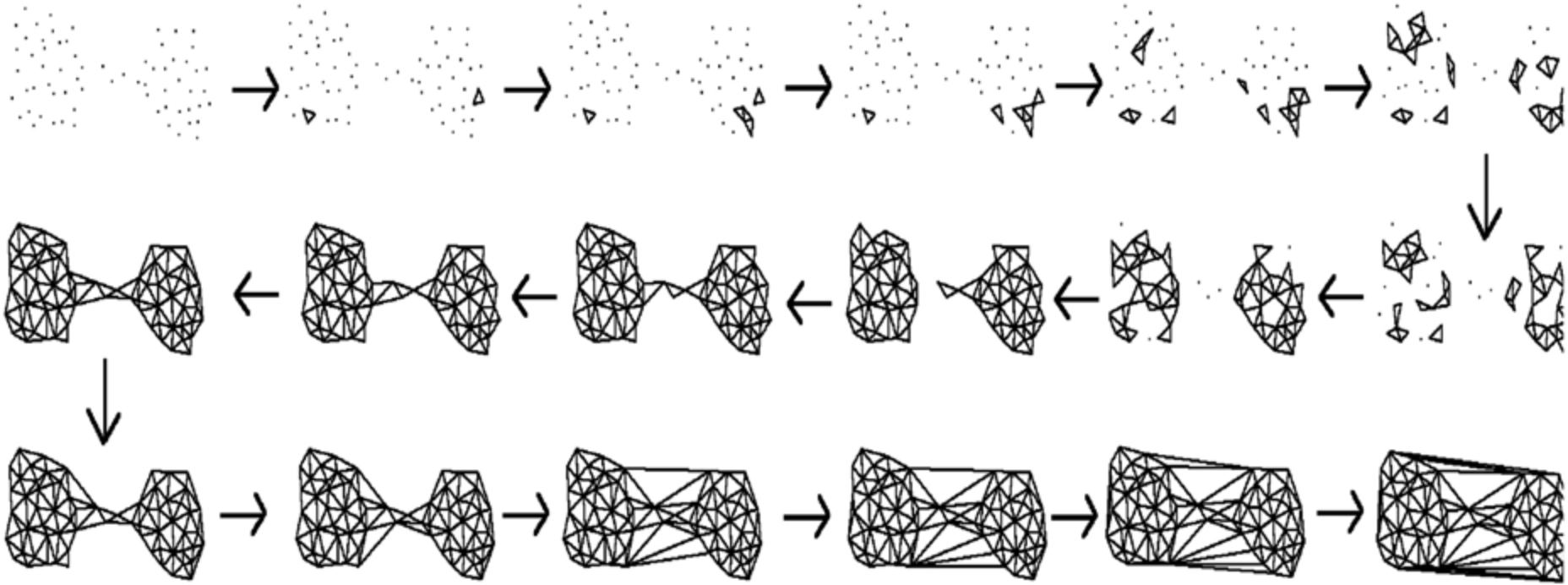
Graph analysis in the different layers of the brain



*Jimenez, G. et al. Unravelling the Topographical Organization of Brain Lesions in Variants of Alzheimer's Disease Progression. SPIE 2025 (rev.)

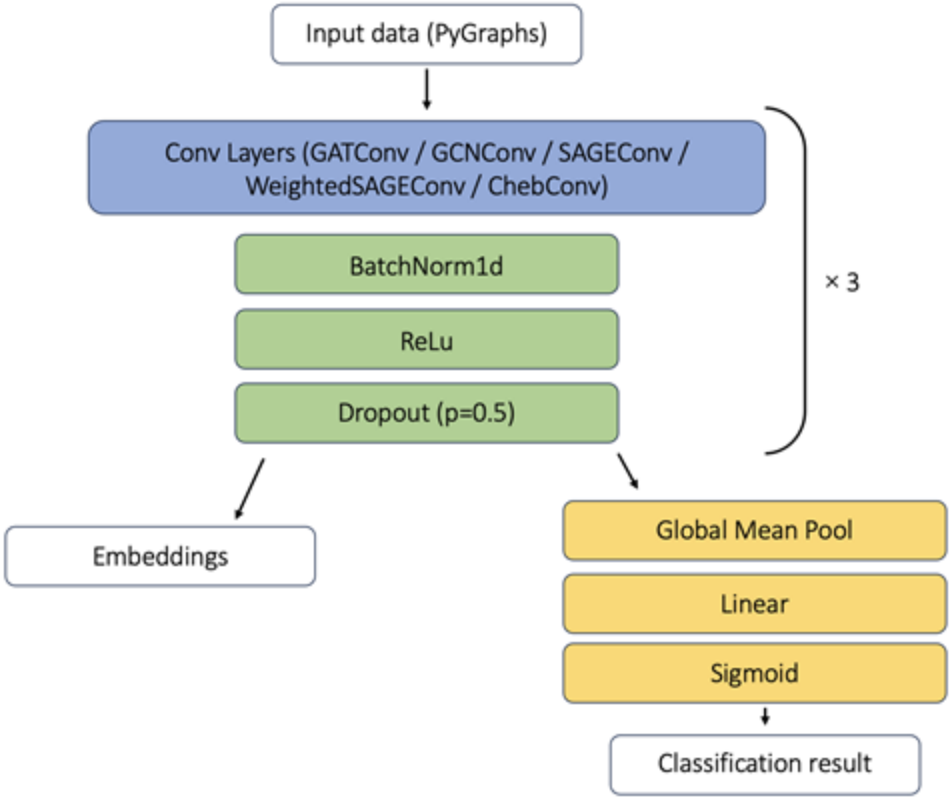
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Graph creation and DL training

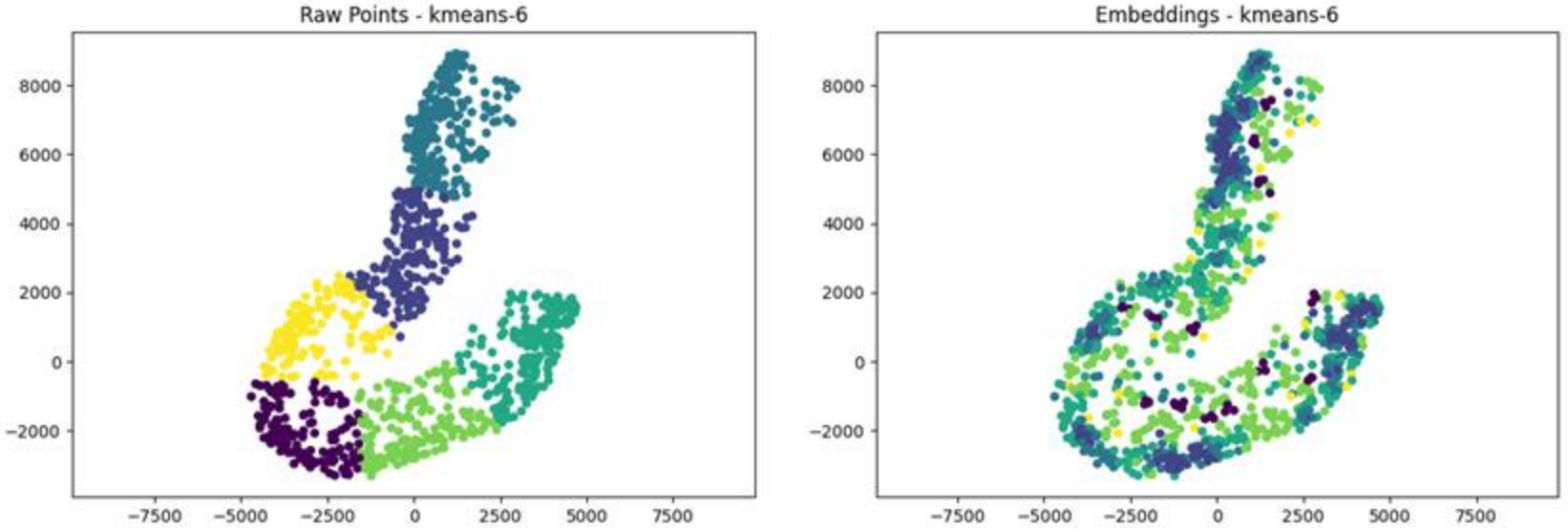


Loménie, Nicolas, and Daniel Racoceanu. 2012. "Point Set Morphological Filtering and Semantic Spatial Configuration Modeling: Application to Microscopic Image and Bio-Structure Analysis." Pattern Recognition 45 (8): 2894–2911.

Graph creation and DL training



Comparison of clustering method for raw data points and GNN-based embeddings. The left graph presents clustered nodes based on their **spatial coordinates**. In contrast, the right graph leverages **GNN embeddings** to cluster nodes.



Cross-validation accuracy and standard deviation of the RF classification between cAD and rpAD classes for plaques and tangles layers.

Layers	1	2	3	4	5	6
Plaques	0.62 ± 0.23	0.72 ± 0.18	0.83 ± 0.16	0.67 ± 0.16	0.62 ± 0.26	0.61 ± 0.23
Tangles	0.57 ± 0.16	0.55 ± 0.29	0.64 ± 0.25	0.72 ± 0.23	0.65 ± 0.22	0.65 ± 0.25

Jimenez, G. et al. Unravelling the Topographical Organization of Brain Lesions in Variants of Alzheimer's Disease Progression. SPIE 2025 (rev.)

- **The key ideas from the study**

1. Understanding the impact of WSI acquisition procedures in DL models.

1. Graphs can integrate local and global information of WSI.

1. GNN and GNN embeddings are a powerful tool to deeply analyze patterns of tangles and plaques present in brain tissue layers.

Conclusion

a. Topological features from graphs allow for AD patient stratification
   - Graph creation only using spatial information

a. High variability of graphs depending on the ROI
   - The study was conducted using manual annotations

a. Graphs and GNN allow the analysis of tangles and plaques in different brain layers
   - Layers were manually annotated following H&E guideline

d. Only spatial relations between nodes were considered.

e. Tangles and plaques were processed independently.



Perspectives

Future extension of the contributions

18/10/2024

AM-FM deconvolution applied to WSI

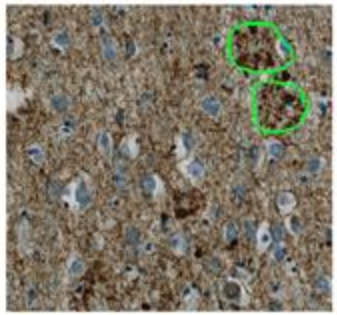
a. Evaluation of the impact of the patch size

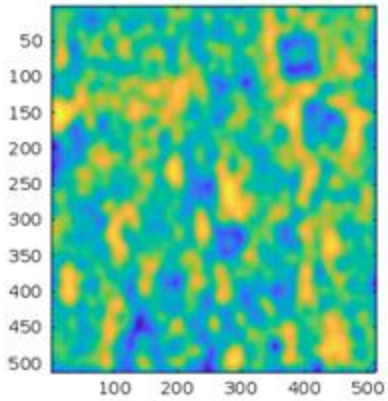
a. Multiscale AM-FM demodulation

- A second or third scale can provide additional textural features due to the complexity of the neuritic plaques

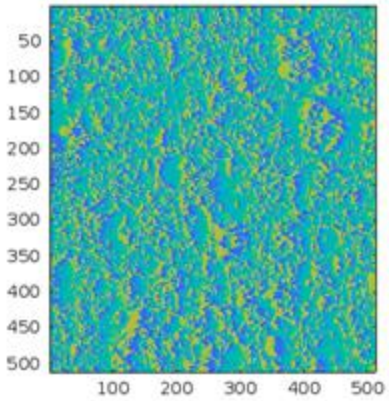
a. Extension to H&E datasets

- H&E focuses on morphological patterns in the tissue





Instantaneous Amplitude



Instantaneous Phase

Graph representation of WSI

a. Relationship between tangles and plaques in the tissue
   - Integration of both graphs as different layers to the GNN

a. Morphological features as node attributes
   - Integrate the morphological features computed from plaques and tangle annotations

a. Potential extension to other diseases related to tauopathies
   - Frontotemporal dementia, CBD, Pick's disease

d. Studies are performed post-mortem.
   - Multimodal image processing. Relationship with aB-PET.



Thanks for your attention

18/10/2024